

Execution screenshot of the queried output

1. DISTINCT VALUES:

Salary

60000
80000
45000
70000
55000
65000
90000
95000
75000
85000
100000

2. Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

Employee_Age Employee_Salary

27	60000
31	80000
26	45000
29	70000
25	55000
28	65000
32	90000
27	70000
30	55000
29	80000
28	60000
26	95000
33	75000
30	60000
27	85000
31	100000
28	80000
25	75000

Employee_Age Employee_Salary

29	55000
32	90000
26	85000
28	60000
30	70000
27	95000
31	100000
29	75000
32	60000
26	60000
33	80000
28	90000

Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Department_ID	Location_ID	Salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000
5004	Kiara Malhotra	F	29	2015-07-05	NULL (or Data Scientist if updated)	8	3	70000
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000

Find the employee whose designation is missing and fill it with "Data Scientist".

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Department_ID	Location_ID	Salary
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000

Find employees sorted by department ID in ascending order and salary in descending order.

Employee_ID	Employee_Name	Department_ID	Salary
5020	Aarav Nair	1	90000
5007	Anushka Singh	2	90000
5023	Anika Paul	2	70000
5011	Saanvi Patel	2	60000
5025	Avni Iyengar	3	100000
5026	Vivaan Singh	3	75000
5001	Vihaan Singh	3	60000
5016	Zoya Khan	4	100000
5022	Aadhya Iyengar	4	60000
5019	Ishika Patel	4	55000
5024	Aryan Shetty	5	95000
5030	Sara Iyer	5	90000
5027	Ananya Paul	6	60000
5028	Anaya Kapoor	6	60000
5015	Ishaan Kumar	7	85000
5021	Advik Kapoor	7	85000
5004	Kiara Malhotra	8	70000
5008	Diya Jha	8	70000
5006	Dhruv Shetty	8	65000
5018	Ishan Mishra	9	75000
5013	Arnav Rao	10	75000
5014	Vihaan Mohan	10	60000
5003	Aaradhya Iyer	10	45000
5005	Anvi Chaudhary	11	55000
5009	Kiaan Desai	11	55000
5002	Reyansh Singh	12	80000
5010	Atharv Yadav	12	80000
5017	Kabir Nair	12	80000
5029	Arjun Kumar	12	80000
5012	Myra Verma	13	95000

Display the first 5 employees hired in the year 2018

Employee_ID	Employee_Name	Gender	Age	Hire_Date	Designation	Department_ID	Location_ID	Salary
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000

Calculate the sum of all salaries in the Finance department.

Employee_ID	Employee_Name	Salary
5015	Ishaan Kumar	85000
5021	Advik Kapoor	85000

Find the minimum age among all employees.

Employee_ID	Employee_Name	Age
5005	Anvi Chaudhary	25
5018	Ishan Mishra	25

List the maximum salary for each location

Location	Maximum_Salary
Chennai	95000
Bangalore	95000
Hyderabad	100000
Pune	100000

Calculate the average salary for each designation containing the word 'Analyst'.

Designation	Average_Salary
Data Analyst	60000
Marketing Analyst	60000
Supply Chain Analyst	60000

Designation	Average_Salary
Financial Analyst	85000
Finance Analyst	85000
Business Analyst	75000
Quality Assurance Analyst	80000

Find departments with less than 3 employees

Department_ID	Department_Name	Employee_Count
1	Software Development	1
5	Product Management	2
6	Content Creation	2
7	Finance	2
9	Research and Development	1
10	Customer Support	2
11	Business Development	2
13	Operations	1

Find locations with female employees whose average age is below 30

Location	Average_Age
Chennai	27.75
Bangalore	27.33
Pune	28.20

List employee names, their designations, and department names where employees are assigned to a department.

Employee Name	Designation	Department Name
Vihaan Singh	Data Analyst	Data Science
Reyansh Singh	Network Engineer	IT
Aaradhya Iyer	Customer Support Executive	Customer Support
Kiara Malhotra	Data Scientist*	Design
Anvi Chaudhary	Business Development Executive	Business Development
Dhruv Shetty	UI Developer	Design
Anushka Singh	Marketing Manager	Marketing
Diya Jha	Graphic Designer	Design
Kiaan Desai	Sales Executive	Business Development
Atharv Yadav	Systems Administrator	IT
Saanvi Patel	Marketing Analyst	Marketing
Myra Verma	Operations Manager	Operations
Arnav Rao	Customer Success Manager	Customer Support

Employee Name	Designation	Department Name
Vihaan Mohan	Supply Chain Analyst	Customer Support
Ishaan Kumar	Financial Analyst	Finance
Zoya Khan	Legal Counsel	Human Resources
Kabir Nair	IT Support Specialist	IT
Ishan Mishra	Research Scientist	Research and Development
Ishika Patel	Talent Acquisition Specialist	Human Resources
Aarav Nair	Software Engineer	Software Development
Advik Kapoor	Finance Analyst	Finance
Aadhya Iyengar	HR Specialist	Human Resources
Anika Paul	Public Relations Specialist	Marketing
Aryan Shetty	Product Manager	Product Management
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Arjun Kumar	Quality Assurance Analyst	IT
Sara Iyer	Project Manager	Product Management

List all departments along with the total number of employees in each Department, including departments with no employees

Employee Name	Designation	Department Name
Vihaan Singh	Data Analyst	Data Science
Reyansh Singh	Network Engineer	IT
Aaradhya Iyer	Customer Support Executive	Customer Support
Kiara Malhotra	Data Scientist*	Design
Anvi Chaudhary	Business Development Executive	Business Development
Dhruv Shetty	UI Developer	Design
Anushka Singh	Marketing Manager	Marketing
Diya Jha	Graphic Designer	Design
Kiaan Desai	Sales Executive	Business Development
Atharv Yadav	Systems Administrator	IT
Saanvi Patel	Marketing Analyst	Marketing
Myra Verma	Operations Manager	Operations
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Ishaan Kumar	Financial Analyst	Finance
Zoya Khan	Legal Counsel	Human Resources

Employee Name	Designation	Department Name
Kabir Nair	IT Support Specialist	IT
Ishan Mishra	Research Scientist	Research and Development
Ishika Patel	Talent Acquisition Specialist	Human Resources
Aarav Nair	Software Engineer	Software Development
Advik Kapoor	Finance Analyst	Finance
Aadhya Iyengar	HR Specialist	Human Resources
Anika Paul	Public Relations Specialist	Marketing
Aryan Shetty	Product Manager	Product Management
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Arjun Kumar	Quality Assurance Analyst	IT
Sara Iyer	Project Manager	Product Management

Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name

Location Employee_Name

Bangalore Aaradhya Iyer
Bangalore Anika Paul
Bangalore Arjun Kumar
Bangalore Dhruv Shetty
Bangalore Kabir Nair
Bangalore Myra Verma
Bangalore Vihaan Mohan
Bangalore Vivaan Singh
Chennai Aadya Iyengar
Chennai Aarav Nair
Chennai Anaya Kapoor
Chennai Anvi Chaudhary
Chennai Aryan Shetty
Chennai Reyansh Singh
Chennai Saanvi Patel
Chennai Sara Iyer
Hyderabad Ananya Paul
Hyderabad Anushka Singh
Hyderabad Arnav Rao
Hyderabad Ishan Mishra

Location Employee_Name

Hyderabad Kiaan Desai
Hyderabad Kiara Malhotra
Hyderabad Advik Kapoor
Pune Avni Iyengar

Show all possible combinations of departments and locations.

Department_Name	Location
Business Development	Bangalore
Business Development	Chennai
Business Development	Hyderabad
Business Development	Pune
Content Creation	Bangalore
Content Creation	Chennai
Content Creation	Hyderabad
Content Creation	Pune
Customer Support	Bangalore
Customer Support	Chennai
Customer Support	Hyderabad
Customer Support	Pune
Data Science	Bangalore
Data Science	Chennai
Data Science	Hyderabad
Data Science	Pune
Design	Bangalore
Design	Chennai
Design	Hyderabad
Design	Pune
Finance	Bangalore
Finance	Chennai
Finance	Hyderabad
Finance	Pune
Human Resources	Bangalore
Human Resources	Chennai
Human Resources	Hyderabad
Human Resources	Pune
IT	Bangalore
IT	Chennai
IT	Hyderabad

Department_Name	Location
IT	Pune
Marketing	Bangalore
Marketing	Chennai
Marketing	Hyderabad
Marketing	Pune
Operations	Bangalore
Operations	Chennai
Operations	Hyderabad
Operations	Pune
Product Management	Bangalore
Product Management	Chennai
Product Management	Hyderabad
Product Management	Pune
Research and Development	Bangalore
Research and Development	Chennai
Research and Development	Hyderabad
Research and Development	Pune
Software Development	Bangalore
Software Development	Chennai
Software Development	Hyderabad
Software Development	Pune

Show pairs of employees working in the same department, excluding self-pairs

Employee_1	Employee_2	Department_Name
Vihaan Singh	Avni Iyengar	Data Science
Vihaan Singh	Vivaan Singh	Data Science
Avni Iyengar	Vivaan Singh	Data Science
Anushka Singh	Saanvi Patel	Marketing
Anushka Singh	Anika Paul	Marketing
Saanvi Patel	Anika Paul	Marketing
Zoya Khan	Ishika Patel	Human Resources
Zoya Khan	Aadhya Iyengar	Human Resources
Ishika Patel	Aadhya Iyengar	Human Resources
Aryan Shetty	Sara Iyer	Product Management
Ananya Paul	Anaya Kapoor	Content Creation
Ishaan Kumar	Advik Kapoor	Finance
Kiara Malhotra	Dhruv Shetty	Design
Kiara Malhotra	Diya Jha	Design

Employee_1	Employee_2	Department_Name
Dhruv Shetty	Diya Jha	Design
Aaradhya Iyer	Arnav Rao	Customer Support
Aaradhya Iyer	Vihaan Mohan	Customer Support
Arnav Rao	Vihaan Mohan	Customer Support
Anvi Chaudhary	Kiaan Desai	Business Development
Reyansh Singh	Atharv Yadav	IT
Reyansh Singh	Kabir Nair	IT
Reyansh Singh	Arjun Kumar	IT
Atharv Yadav	Kabir Nair	IT
Atharv Yadav	Arjun Kumar	IT
Kabir Nair	Arjun Kumar	IT

Write a window function query to rank employees by salary using rank(). • Write a window function query to rank employees by salary within each department using DENSE_RANK()

Employee_ID	Employee_Name	Salary	Salary_Rank
5016	Zoya Khan	100000	1
5025	Avni Iyengar	100000	1
5012	Myra Verma	95000	3
5024	Aryan Shetty	95000	3
5007	Anushka Singh	90000	5
5020	Aarav Nair	90000	5
5030	Sara Iyer	90000	5
5015	Ishaan Kumar	85000	8
5021	Advik Kapoor	85000	8
5002	Reyansh Singh	80000	10
5010	Atharv Yadav	80000	10
5017	Kabir Nair	80000	10
5029	Arjun Kumar	80000	10
5013	Arnav Rao	75000	14
5018	Ishan Mishra	75000	14
5026	Vivaan Singh	75000	14
5004	Kiara Malhotra	70000	17
5008	Diya Jha	70000	17
5023	Anika Paul	70000	17
5006	Dhruv Shetty	65000	20
5001	Vihaan Singh	60000	21
5011	Saanvi Patel	60000	21

Employee_ID	Employee_Name	Salary	Salary_Rank
5014	Vihaan Mohan	60000	21
5022	Aadhya Iyengar	60000	21
5027	Ananya Paul	60000	21
5028	Anaya Kapoor	60000	21

Write a window function query, Running total salary by department **Deliverables**

Employee_Name	Department_ID	Salary	Running_Total_Salary
Aarav Nair	1	90000	90000
Saanvi Patel	2	60000	60000
Anika Paul	2	70000	130000
Anushka Singh	2	90000	220000
Vihaan Singh	3	60000	60000
Vivaan Singh	3	75000	135000
Avni Iyengar	3	100000	235000
Ishika Patel	4	55000	55000
Aadhya Iyengar	4	60000	115000
Zoya Khan	4	100000	215000
Sara Iyer	5	90000	90000
Aryan Shetty	5	95000	185000
Ananya Paul	6	60000	60000
Anaya Kapoor	6	60000	120000
Ishaan Kumar	7	85000	85000
Advik Kapoor	7	85000	170000
Dhruv Shetty	8	65000	65000
Kiara Malhotra	8	70000	135000
Diya Jha	8	70000	205000
Ishan Mishra	9	75000	75000
Aaradhya Iyer	10	45000	45000
Vihaan Mohan	10	60000	105000
Arnav Rao	10	75000	180000
Anvi Chaudhary	11	55000	55000
Kiaan Desai	11	55000	110000
Reyansh Singh	12	80000	80000
Atharv Yadav	12	80000	160000
Kabir Nair	12	80000	240000
Arjun Kumar	12	80000	320000
Myra Verma	13	95000	95000

